

Difference-in-Differences with Multiple Time Periods*

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Abstract

In this article, we consider identification, estimation, and inference procedures for treatment effect parameters using Difference-in-Differences (DiD) with (i) multiple time periods, (ii) variation in treatment timing, and (iii) when the “parallel trends assumption” holds potentially only after conditioning on observed covariates. We show that a family of causal effect parameters are identified in staggered DiD setups, even if differences in observed characteristics create non-parallel outcome dynamics between groups. Our identification results allow one to use outcome regression, inverse probability weighting, or doubly-robust estimands. We also propose different aggregation schemes that can be used to highlight treatment effect heterogeneity across different dimensions as well as to summarize the overall effect of participating in the treatment. We establish the asymptotic properties of the proposed estimators and prove the validity of a computationally convenient bootstrap procedure to conduct asymptotically valid simultaneous (instead of pointwise) inference. Finally, we illustrate the relevance of our proposed tools by analyzing the effect of the minimum wage on teen employment from 2001–2007. Open-source software is available for implementing the proposed methods.

JEL: C14, C21, C23, J23, J38.

Keywords: Difference-in-Differences, Dynamic Treatment Effects, Doubly Robust, Event Study, Variation in Treatment Timing, Treatment Effect Heterogeneity, Semi-Parametric.

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1 Introduction

Difference-in-Differences (DiD) has become one of the most popular research designs used to evaluate causal effects of policy interventions. In its canonical format, there are two time periods and two groups: in the first period no one is treated, and in the second period some units are treated (the treated group), and some units are not (the comparison group). If, in the absence of treatment, the average outcomes for treated and comparison groups would have followed parallel paths over time (which is the so-called parallel trends assumption), one can estimate the average treatment effect for the treated subpopulation (ATT) by comparing the average change in outcomes experienced by the treated group to the average change in outcomes experienced by the comparison group. Methodological extensions of DiD methods often focus on this standard two periods, two groups setup; see, e.g., [Heckman et al. \(1997, 1998\)](#), [Abadie \(2005\)](#), [Athey and Imbens \(2006\)](#), [Qin and Zhang \(2008\)](#), [Bonhomme and Sauder \(2011\)](#), [de Chaisemartin and D’Haultfœuille \(2017\)](#), [Botosaru and Gutierrez \(2018\)](#), [Callaway et al. \(2018\)](#), and [Sant’Anna and Zhao \(2020\)](#).¹

Many DiD empirical applications, however, deviate from the canonical DiD setup and have more than two time periods and variation in treatment timing. In this article, we provide a unified framework for average treatment effects in DiD setups with multiple time periods, variation in treatment timing, and when the parallel trends assumption holds potentially only after conditioning on observed covariates. We concentrate our attention on DiD with staggered adoption, i.e., to DiD setups such that once units are treated, they remain treated in the following periods.

The core of our proposal relies on separating the DiD analysis into three separate steps: (i) identification of policy-relevant disaggregated causal parameters; (ii) aggregation of these parameters to form summary measures of the causal effects; and (iii) estimation and inference about these different target parameters. Our approach allows for estimation and inference on interpretable causal parameters allowing for arbitrary treatment effect heterogeneity and dynamic effects, thereby completely avoiding the issues of interpreting results of standard two-way fixed effects (TWFE) regressions as causal effects in DiD setups as pointed out by [Borusyak and Jaravel \(2017\)](#), [de Chaisemartin and D’Haultfœuille \(2020\)](#), [Goodman-Bacon \(2019\)](#), [Sun and Abraham \(2020\)](#), and [Athey and Imbens \(2018\)](#). In addition, it adds transparency and objectivity to the analysis ([Rubin \(2007, 2008\)](#)), and allows researchers to exploit a variety of estimation methods to answer different questions of interest.

The identification step of the analysis provides a blueprint for the other steps. In this paper, we pay particular attention to the disaggregated causal parameter that we call the *group-time average treatment effect*, i.e., the average treatment effect for group g at time t , where a “group” is defined by the time period when units are first treated. In the canonical DiD setup with two periods and two groups, these parameters reduce to the ATT which is typically the parameter of interest in that setup. An attractive feature of the group-time average treatment effect parameters is that they do not directly restrict heterogeneity with respect to observed covariates, the period in which units are first treated, or the evolution of treatment effects over time. As a consequence, these easy-to-interpret causal parameters can be directly used for learning about treatment effect heterogeneity, and/or to construct many other more aggregated causal

¹See Section 6 of [Athey and Imbens \(2006\)](#) and Theorem S1 in [de Chaisemartin and D’Haultfœuille \(2017\)](#) for notable exceptions that cover multiple periods and multiple groups.

These expressions for $ATT(g, t)$ clearly resemble the one for ATT in the canonical two-periods and two-groups case. As in that case, the average effect of participating in the treatment for units in group g is identified by taking the path of outcomes (i.e., the change in outcomes between the most recent period before they were affected by the treatment and the current period) actually experienced by that group and adjusting it by the path of outcomes experienced by a comparison group. Under the parallel trends assumption, this latter path is the path of outcomes that units in group g would have experienced if they had not participated in the treatment.

Remark 3. From (2.8) one can see that when Assumptions 3 and 4 hold unconditionally and there is no-anticipation, the $ATT(g, t)$ parameter can be obtained by first subsetting the data to only contain observations at time t and $g - 1$, from units with either $G_g = 1$ or $C = 1$, and then, using only the observations of this subset, running the (population) linear regression

$$Y = \alpha_1^{g,t} + \alpha_2^{g,t} \cdot G_g + \alpha_3^{g,t} \cdot 1\{T = t\} + \beta^{g,t} \cdot (G_g \times 1\{T = t\}) + \epsilon^{g,t}. \quad (2.10)$$

It is then easy to verify that $\beta^{g,t} = ATT(g, t)$. Note that one would need to consider different partitions of the data to characterize different $ATT(g, t)$ in terms of regression parameters. Alternatively, one could use the interacted two-way fixed effects regression proposed by [Sun and Abraham \(2020\)](#).

Remark 4. When covariates are available, the $\tilde{\beta}^{g,t}$ coefficient of the population linear regression

$$Y = \tilde{\alpha}_1^{g,t} + \tilde{\alpha}_2^{g,t} \cdot G_g + \tilde{\alpha}_3^{g,t} \cdot 1\{T = t\} + \tilde{\beta}^{g,t} \cdot (G_g \times 1\{T = t\}) + \tilde{\gamma} \cdot X + \tilde{\epsilon}^{g,t}$$

that uses the same subset of data as in Remark 3 is, in general, not equal to $ATT(g, t)$ unless one is willing to assume (i) homogeneous (in X) treatment effects, i.e., $\mathbb{E}[Y_t(g) - Y_t(0)|G_g = 1, X] = \mathbb{E}[Y_t(g) - Y_t(0)|G_g = 1]$ a.s., and (ii) rule-out covariate-specific trends, i.e., for $\mathbb{E}[Y_t - Y_{t-1}|X, G] = \mathbb{E}[Y_t - Y_{t-1}|G]$ a.s. for all groups and time periods; see, e.g., [Śłoczyński \(2018\)](#) for a related discussion. The characterizations of $ATT(g, t)$ discussed in Theorem 1 do not rely on these restrictive conditions.

Remark 5. Although the IPW, OR, and DR based estimands presented in Theorem 1 are identical from an identification standpoint, this is not the case when one wants to estimate and make inference about the $ATT(g, t)$. As we discuss in Section 4, DiD estimators based on the DR estimands (2.4) and (2.7) usually enjoy additional robustness against model-misspecifications when compared to the IPW and OR estimands.

Remark 6. Theorem 1 suggests that we can identify $ATT(g, t)$ only for groups in $\mathcal{G}_\delta \subseteq \mathcal{G}$ which can involve dropping some “early treated” groups due to anticipation effects. When $\delta = 0$, i.e. when there is no anticipation, $\mathcal{G}_\delta = \mathcal{G}$. Theorem 1 also suggests that we can identify $ATT(g, t)$ only until $t = \mathcal{T} - \delta$ because of potential treatment anticipation behavior. In applications where some units are known to never participate in the treatment (including periods after time period $\mathcal{T}$), however, we can identify $ATT(g, t)$ up to $t = \mathcal{T}$ by using these units as a valid comparison group for all time periods $t = \mathcal{T} - \delta + 1, \dots, \mathcal{T}$, provided that an appropriate parallel trends assumption is satisfied.

Remark 7. From Theorem 1 it is clear that pre-treatment covariates play a prominent role in our analysis. Importantly, Assumptions 4 and 5 suggest that researchers should include pre-treatment covariates

all event times or $\theta_c(t)$ across all time periods, i.e.,

$$\theta_{es}^O = \frac{1}{\mathcal{T}-1} \sum_{e=0}^{\mathcal{T}-2} \theta_{es}(e) \quad \theta_c^O = \frac{1}{\mathcal{T}-1} \sum_{t=2}^{\mathcal{T}} \theta_c(t) \quad (3.12)$$

In our view, the appeal of these aggregations is likely to be somewhat more limited than that of θ_{sel}^O in most applications. For example, the interpretation of θ_{es}^O is complicated by the issue of the changing composition of groups across different values of e discussed above (similar arguments apply to θ_c^O as well). As before, one can circumvent the issue of the changing composition of groups by balancing the sample with respect to event time. A (local) single summary parameter is given by

$$\theta_{es}^{O,bal}(e') = \frac{1}{e'+1} \sum_{e=0}^{e'} \theta_{es}^{bal}(e, e') \quad (3.13)$$

This is the average effect of participating in the treatment over the first e' periods of exposure to the treatment. This is also a reasonable alternative overall treatment effect parameter, but it should also be noted that it is local to groups that participated in the treatment for at least e' periods.

As a final comment, in general, none of the overall effect parameters considered in this section are equal to each other except in the special case where $ATT(g, t)$ is the same for all groups and all time periods. In that case, all of the aggregated parameters, including β from the TWFE regression, are equal to each other.

4 Estimation and Inference

So far we have focused on the identification and aggregation stages of the analysis. In this section, we show how one can build on these results to form estimators for and conduct inference about the group-time average treatment effects and their summary measures described in Section 3. Given that the $ATT(g, t)$'s are the main building blocks of our analysis, we start with them.

First, it is important to notice that our identification results in Theorem 1 are constructive and suggest a simple and intuitive two-step estimation strategy to estimate the $ATT(g, t)$'s. In the first step, one estimates the nuisance functions for each group g and time period t — $p_g(x)$ and/or $m_{g,t,\delta}^{nev}(X)$ if one relies on Assumption 4, and $p_{g,t+\delta}(x)$ and/or $m_{g,t,\delta}^{ny}(X)$ if one relies on Assumption 5. In the second step, one plugs the fitted values of these estimated nuisance functions into the sample analogue of the considered $ATT(g, t)$ estimand to obtain estimates of the group-time average treatment effect.

A natural question that then arises is which type of approach one should use in practice: the outcome regression, inverse probability weighting, or the doubly-robust one. Although these three different approaches are equivalent from the *identification* perspective, this is not the case from the *estimation/inference* perspective. The OR approach requires researchers to *correctly* model the outcome evolution of the comparison group to estimate the group-time average treatment effects. This approach is explicitly connected with the conditional parallel trends assumption required in DiD analysis as this condition is usually expressed in terms of conditional expectations. The IPW approach, on the other hand, avoids explicitly modeling the outcome evolution of the comparison group and therefore does not